

AKASH KARMAKAR

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Education

Baruch College, Zicklin School of Business, MS in Finance-Fintech - New York Aug 2024 – May 2026

- GPA: 3.6/4.0 , **Coursework:** Quantitative Investments, Programming, Machine Learning in Investment

Jadavpur University, Bachelor in Engineering - Kolkata July 2018 – July 2022

- GPA: 3.7/4.0 , **Coursework:** Machine Learning, Mathematics 1 & 11, Numerical Methods

Experience

Quantitative Research Consultant, WorldQuant, US April 2025 – June 2025

- Designed and simulated trading signals across US, EMEA and APAC markets, achieving exceptional performance metrics, including a portfolio Sharpe ratio of 2.24 and a RoMaD of 9.21 in simulation.
- Optimized market-neutral portfolios and created alphas delivering annualized returns exceeding 15% with low drawdowns, validated through multi-regime analysis, parameter sensitivity tests and robustness checks to ensure stability and minimize overfitting.

Software Engineer, Goprac – Bengaluru, India June 2023 – April 2024

- Automated the interview question upload flow with auto-validation and cluster balancing, making uploads faster, safer and more accurate while cutting labor time and cost.
- Collaborated with the product team and led the integration of an AI-driven personalized feedback system using skills-based scoring, optimizing assessment workflows, reducing friction by removing manual review and improving efficiency for 36,000+ users.
- Engineered and implemented database procedures, functions and API integrations to automate server-side operations, streamlining the architecture and reducing processing and debugging time.

Data Scientist, Celebal Technologies – Jaipur, India July 2022 – June 2023

- Designed prompts using Chain-of-Thought for Text2SQL, Multi-intent classification and medical report analysis.
- Assessed advanced data mining techniques and conducted exploratory data analysis (EDA) on unstructured data, leading to significant insights for feature engineering.
- Analyzed and executed algorithms, K-Means Clustering, Regression to extract meaningful information from data.

Technical Skills

Languages/Frameworks: C , C++ , Python , Java , Javascript , Node.js , React.js , SQL , PHP , Rust , Tensorflow

Technologies: Postman , Linux , NLP , AWS , MongoDB , SQL Server , XCode , Redis , Power BI , Excel , OpenAI

Achievements

- Achieved US rank 3 and global rank 14 out of 76,867 in the IQC worldQuant 2025 (National Finalist).
- Secured 3rd place in the Zicklin Graduate Entrepreneurship Competition 2025.
- Ranked globally 1056th in Google Kickstart 2021 coding competition round E.

Projects

Dynamic Sector Rotation with Machine Learning [Github link](#)

- Designed and implemented a LightGBM-based sector rotation strategy integrating PCA and K-Means clustering for market regime detection and ETF allocation optimization, achieving a Sharpe ratio of 1.12 in-sample. Conducted robust backtesting across 2014–2025 market regimes, outperforming the Pairs Portfolio benchmark in out-of-sample returns (8.76% vs. 7.35%) while controlling drawdowns.

Portfolio Optimization and Efficient-Frontier Visualizer [Github link](#)

- Portfolio optimizer leveraging historical stock data to simulate 10,000 portfolios and visualize efficient frontier.

Mern Ecommerce Website (Node.js , React.js , Redux , MongoDB) [Website link](#)

- Full-stack e-commerce website with JWT authentication, search functionality and a user-friendly cart system.